

NMOSFET Design Project

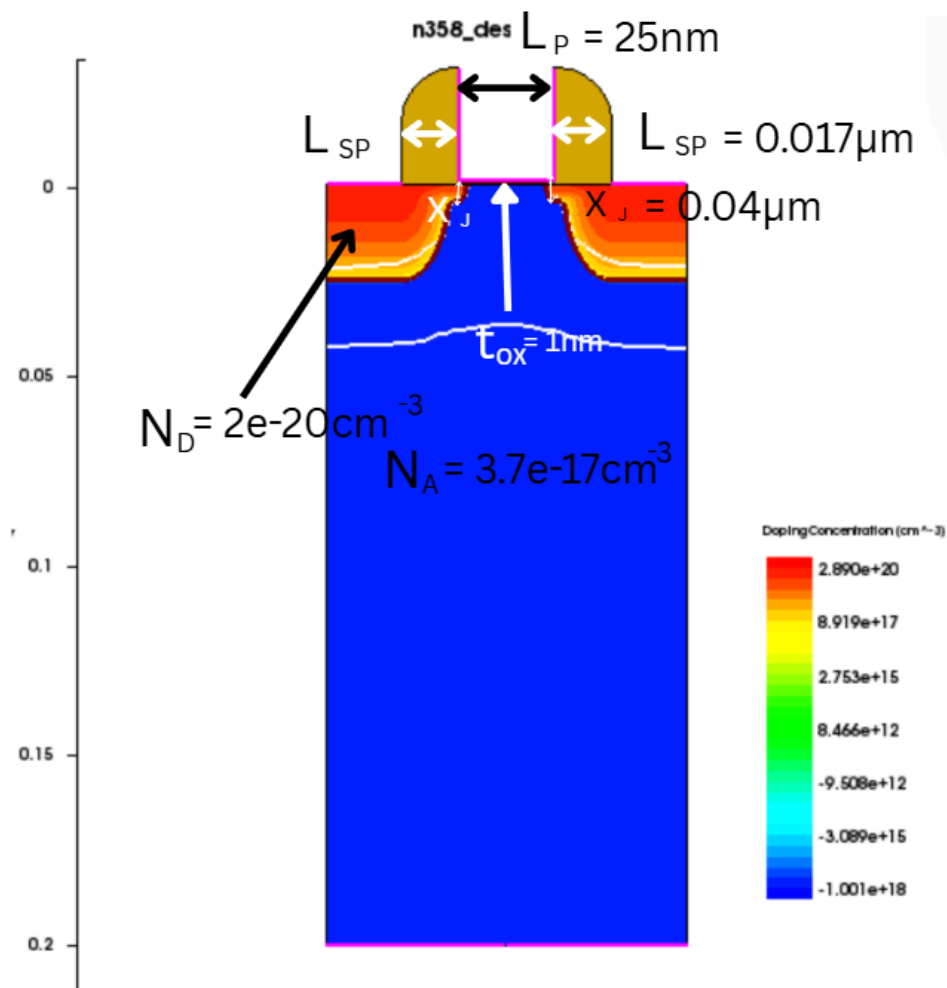
By

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and

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1. [20 pts] Illustrate your MOSFET design, labeling and indicating your values for the design parameters. Describe the process (in a paragraph) by which you arrived at your final design.



The process by which we achieved the target values for ION and IOFF began with a theory-driven initial guess. Based on device physics and the given design constraints, we knew:

- A. Increasing the substrate doping concentration N_A raises the threshold voltage V_T , which reduces IOFF by requiring a higher gate voltage to turn on the transistor. However, a

higher NA also increases impurity scattering, degrading carrier mobility and thereby lowering ION.

- B. A shallower XJ (junction depth) suppresses short-channel effects and lowers IOFF, but it also increases parasitic series resistance, which reduces ION.
- C. Longer spacer lengths reduce the impact of the deep source/drain regions on the channel, improving electrostatic control, but they also increase parasitic series resistance.

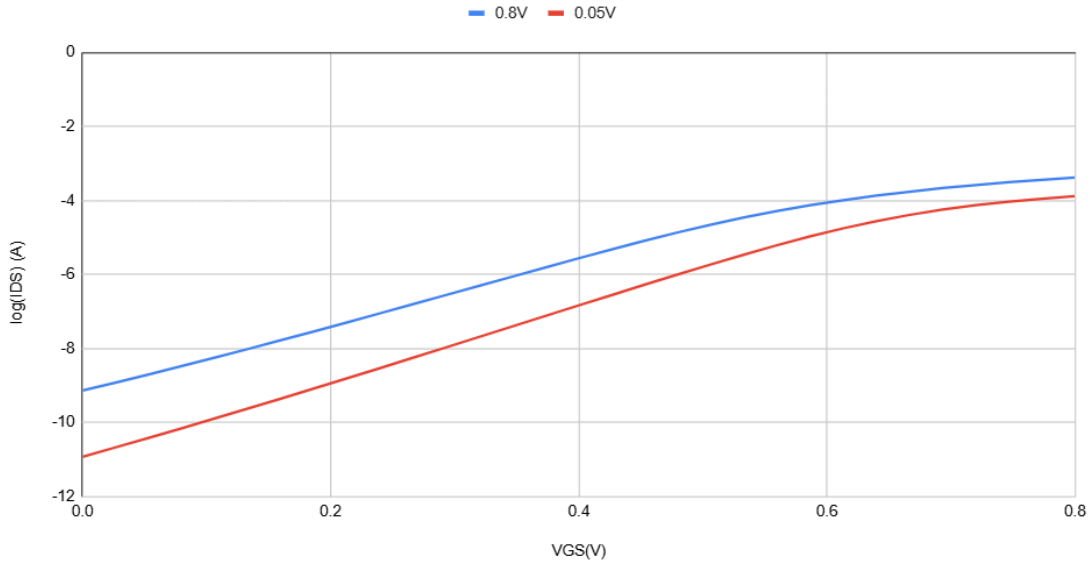
While our first parameter choices didn't meet both current targets simultaneously, we systematically tuned each variable while observing its impact on key figures of merit. Through this iterative exploration, we found that setting $LSP = 0.015 \text{ } \mu\text{m}$, $NA = 1.0 \times 10^{18} \text{ cm}^{-3}$, and $XJ = 0.004 \text{ } \mu\text{m}$ provided the best trade-off. This configuration produced a strong ION of $415 \text{ } \mu\text{A}/\mu\text{m}$, while keeping IOFF at $726 \text{ pA}/\mu\text{m}$ — well below the $1 \text{ nA}/\mu\text{m}$ threshold.

Our overall design goal was to realize a high-performance N-channel MOSFET optimized for speed and power efficiency. Given the fixed gate length of 25 nm , 1 nm oxide thickness, and a metal gate work function of 4.6 eV , we carefully balanced the doping and structural parameters to ensure strong drive current without compromising off-state leakage. In the end, the key trade-off was between suppressing leakage and preserving drive strength. By carefully managing the competing effects of doping, junction engineering, and spacer design, our final device meets the performance targets.

2. [20 pts] Run the IDS-VGS simulation for $V_{DS} = 50 \text{ mV}$ (corresponding to the linear region of operation) and for $V_{DS} = V_{DD}$.

IDS-VGS

$V_{GS} = V_{DD}$, $V_{DS} = 0.05 \text{ V}$



- **Minimum subthreshold swing**

We found the minimum subthreshold swing to be approximately 60 mV/decade. By finding the slope we got around 59mV/decade and by using the equation below we got 60.9mV/decade.

$$S = \frac{\Delta V_G}{\text{decade } I_D} = (\ln 10) \frac{dV_G}{d \ln I_D} = (\ln 10) \left(\frac{kT}{q} \right) \left(\frac{C_{ox} + C_D}{C_{ox}} \right)$$

- **Linear threshold voltage**

The linear threshold voltage we extracted was: **VTlin** = 0.621V.

The saturation threshold voltage we extracted was: **VtSat** = 0.448

- **DIBL** = $\frac{V_{Tlin} - V_t}{V_{DD} - V_{Dlin}} = 161.3333333 \text{ mV/V}$

- $V_{T,lin} = 0.540 \text{ V}$
- $V_{T,sat} = 0.419 \text{ V}$
- $V_{DD} = 0.8 \text{ V}$
- $V_{DS,lin} = 0.05 \text{ V}$

3. [10 pts] What is the threshold voltage (V_T) calculated using the MOSCAP theory as mentioned in the textbook?

The threshold voltage calculated using the MOSCAP theory is based on a simplified, idealized view of the MOS structure.

Voltage Threshold using MOSCAP Theory: $V_T = 1.32\text{V}$

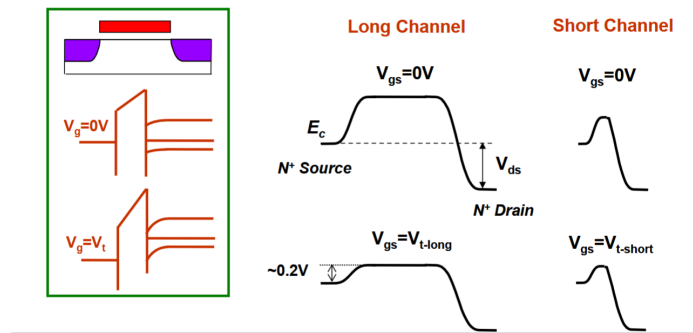
In this method, the MOS capacitor is analyzed in equilibrium (no current flowing). V_T is defined as the gate voltage at which inversion begins — which is when the surface potential reaches twice the Fermi potential ($\phi_s = 2\phi_F$). This calculation includes only static variables: oxide thickness, channel doping (N_A), and gate work function difference, and it also assumes an infinitely long channel, no drain bias, and perfect interfaces.

However, the V_{Tlin} and V_{Tsat} values we extracted from our simulation are different because they reflect real device behavior under bias — including V_{DS} , short-channel effects, and dynamic current conduction. More specifically, the simulations we ran account for:

- Drain-induced barrier lowering (DIBL) — where a high drain voltage reduces the source-channel barrier, effectively lowering V_T
- Non-uniform doping profiles in the extension regions (SDE)
- Finite channel length, which reduces electrostatic control from the gate
- Potential quantum tunneling

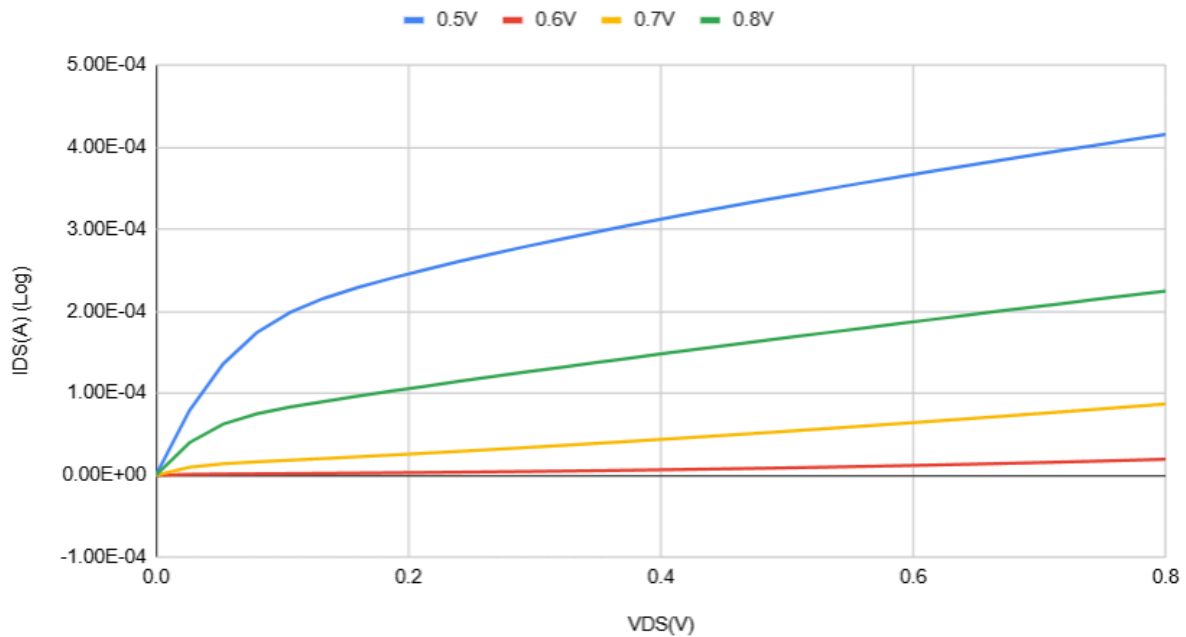
As a result, the simulated V_{Tlin} and V_{Tsat} we found are typically lower than the MOSCAP-calculated V_T , and this discrepancy grows with shorter channels.

This is illustrated in the barrier diagram (shown in lecture): a long-channel device maintains a strong barrier even at $V_{DS} > 0$, while a short-channel device exhibits barrier lowering — shifting the onset of inversion to a lower gate voltage.



4. [20 pts] Run IDS-VDS simulations for $V_{GS} = 0.5, 0.6, 0.7,$ and 0.8 V and plot the IDS-VDS.

IDS-VDS



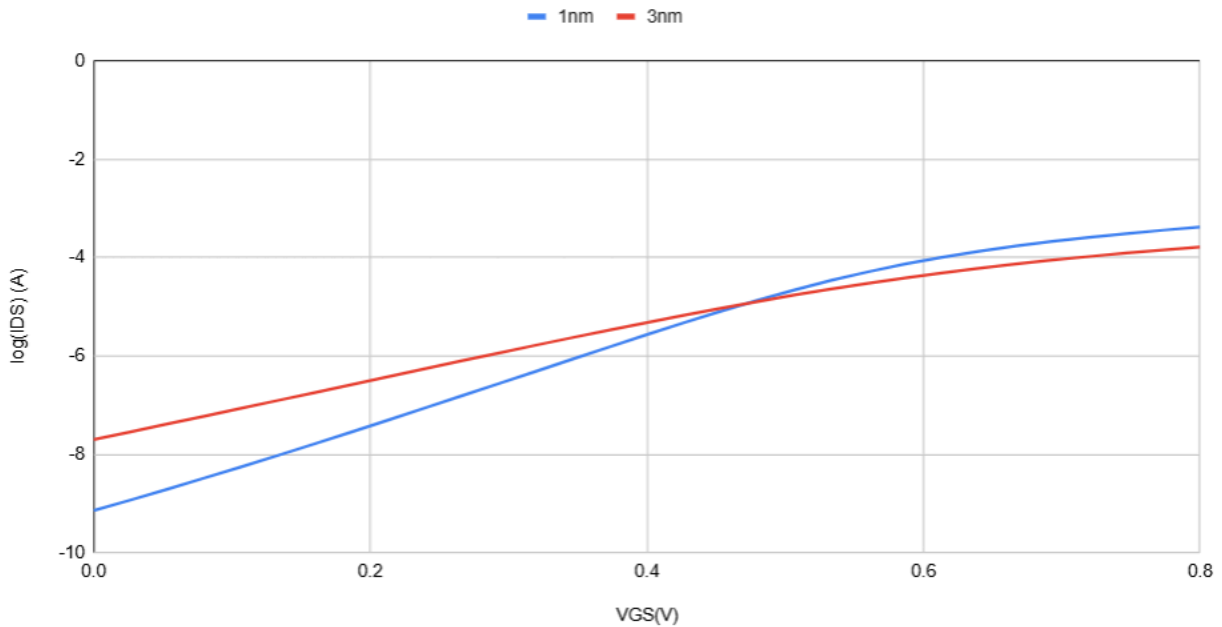
Because of nonideal effects like short-channel behavior and DIBL, which lowers the threshold voltage, the I_{DS} vs. V_{GS} curves do not display square-law dependence of saturation current on V_{GS} which is evident in the curve's shape.

In short-channel MOSFETs, increasing V_{GS} highlights nonideal effects like velocity saturation and channel length modulation, both of which cause deviations from the ideal square-law behavior. At low V_{GS} , I_{DS} appears more linear, indicating the saturation model doesn't fully capture device behavior, likely due to short-channel effects. As V_{GS} increases, I_{DS} continues to rise even in saturation, driven by these nonidealities.

5. [10 pts] Change the effective oxide thickness (EOT) to 3 nm and comment on how that changes the I_{DS} - V_{GS} at $V_{DS} = V_{DD}$. Please comment on how oxide thickness will affect the future generations when gate length is further reduced?

IDS-VGS

tox = 1nm, tox = 3nm



The 1 nm oxide device (blue curve) has a steeper subthreshold slope, meaning it switches more abruptly and achieves higher I_{DS} at lower V_{GS} , indicating better electrostatic control and a lower subthreshold swing. However, this also means higher gate capacitance (C_{ox}) and stronger drive current, which leads to greater dynamic power consumption since switching energy scales with $C_{ox}V_{GS}^2$.

So while the thin oxide improves performance and reduces I_{OFF} , it also makes the transistor more power-hungry compared to $t_{ox} = 3\text{nm}$.

6. [10 pts] If we'd like to reduce OFF current significantly without giving up ON current too much, i.e., increase the ON/OFF ratio while keeping the ON current almost intact, then, what design modifications would you suggest? Justify your answer. For this question, you are allowed to modify the parameters that were fixed earlier (listed under the heading 'Fixed Design Parameters'), except for the channel length, which still has to be 25 nm.

To improve the ON/OFF ratio without sacrificing too much ON current, we would suggest the following design changes (within the allowed parameter range):

Reduce junction depth (XJ): A shallower junction helps suppress short-channel effects and improves gate control near the source. This reduces drain-induced barrier lowering (DIBL), which in turn lowers I_{OFF}. Additionally, reducing XJ helps minimize leakage through junction tunneling.

We could also slightly increase body doping: A modest increase in N_A raises the threshold voltage (V_T), which helps reduce subthreshold leakage. It also tightens the depletion width, improving electrostatics. However, the increase should be moderate to avoid degrading carrier mobility, which would reduce I_{ON}.

As mentioned before, we could reduce effective oxide thickness (EOT): Lowering the oxide thickness increases the gate oxide capacitance, improving electrostatic control and reducing DIBL. This results in a sharper subthreshold slope and lower I_{OFF}. This change also helps preserve I_{ON} by maintaining strong inversion even with slightly higher V_T.

Another idea, though less feasible, might be to increase gate work function (if allowed, would mean no longer using titanium nitride). Choosing a gate material with a higher work function increases V_T , which can help suppress IOFF. This could be helpful if we didn't want to change channel length.

7. [10 pts] How would you expect the ON current, OFF current, and subthreshold swing to change with increasing temperature (e.g. to 85 degrees Celsius)? Briefly explain.

As temperature increases: First, carrier mobility decreases due to enhanced phonon scattering, which tends to reduce I_{ON} slightly. At the same time, the intrinsic carrier concentration increases exponentially, which has a strong impact on I_{OFF} and subthreshold swing.

Specifically, since subthreshold swing (S) is proportional to temperature:

$$S = \ln(10) \cdot kTq \cdot (1 + C_{dep}/C_{ox}),$$

it increases with rising temperature. Thus, we expect that a higher subthreshold swing causes the device to transition more gradually from OFF to ON, allowing more leakage current below threshold. We would see therefore that I_{OFF} increases significantly, and the device becomes less energy efficient in the subthreshold region.

From office hours, it was noted that I_{OFF} is very sensitive to electrostatic control, as seen in its strong dependence on LSP. This sensitivity is further exacerbated at higher temperatures, making I_{OFF} more difficult to control. In summary, increasing temperature generally leads to a slight decrease in I_{ON} , a large increase in I_{OFF} , and an increase in subthreshold swing.